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# Improving Eczema Severity Classification through Attention-Enhanced Deep Learning Pipelines

*Final Project Report*

CSCI 5922: Deep Neural Networks

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## 1 Introduction

Eczema (atopic dermatitis) affects roughly 230 million people worldwide and is one of the most common chronic inflammatory skin conditions [1]. Severity assessment plays a central role in treatment decisions: mild cases are typically managed with topical emollients, while severe cases may require systemic therapy or biologics. Clinicians currently rely on manual scoring systems such as SCORAD and EASI, which are time-consuming and suffer from significant inter-rater variability [2].

Deep learning has shown strong results for detecting whether eczema is present, often exceeding 95% accuracy [3, 4]. However, classifying *how severe* it is (mild vs. moderate vs. severe) remains a much harder and largely unsolved problem. Most existing work treats skin disease classification as a binary or coarse-grained task, while fine-grained severity grading has received relatively little attention in the literature [5].

In this project, we propose a three-stage CNN pipeline for eczema severity classification. The pipeline sequentially (1) validates whether an image contains skin, (2) detects whether eczema is present, and (3) classifies severity into mild, moderate, or severe. We start with a baseline using MobileNetV2 [6] as a feature extractor, then improve the severity stage by incorporating the Convolutional Block Attention Module (CBAM) [7], a stronger backbone (EfficientNetB0) [8], data augmentation, and Grad-CAM++ explainability [11].

## 2 Related Work

**CNN-Based Skin Disease Classification.** Velasco et al. [3] used MobileNet to classify seven skin diseases from smartphone images at 94.4% accuracy. Alenezi et al. [4] combined MobileNetV2 with LSTM for multi-class classification on DermNet. Wu et al. [12] showed that fusing patient metadata with image features further improves diagnostic performance. These works focus on disease-level classification rather than severity grading, which is the gap we aim to address.

**Attention Mechanisms in Medical Imaging.** Woo et al. [7] proposed CBAM, which sequentially applies channel and spatial attention to refine feature maps. Schlemper et al. [13] demonstrated that attention gating

helps localize pathological structures in medical image segmentation. Zhang et al. [14] applied attention-based deep metric learning to dermoscopy images and reported improved inter-class separation. These results suggest attention mechanisms are well suited for fine-grained visual tasks in dermatology.

**Explainability in Dermatological AI.** Selvaraju et al. [10] introduced Grad-CAM for producing visual explanations of CNN decisions. Barata et al. [15] reviewed deep learning for skin diagnosis and found most systems lack clinical validation of their visual explanations. Rezk et al. [16] combined CNN classification with Grad-CAM overlays on HAM10000 and achieved 91.15% accuracy alongside interpretable heatmaps.

**Eczema Severity Scoring with AI.** Hurault et al. [2] developed an automated SCORAD scoring system and highlighted inter-rater variability as a fundamental bottleneck. Cho et al. [5] compared several CNN architectures for atopic dermatitis severity classification. Pan et al. [17] explored transformer-based models for skin disease classification on DermNet. Despite these efforts, robust severity grading from images alone remains an open challenge, which motivates the present project.

## 3 Methods

Our approach is a three-stage CNN pipeline where each stage handles a progressively more specific task. We first build a baseline using transfer learning with MobileNetV2, then improve the severity stage with architectural and data-level changes.

### 3.1 Pipeline Overview

The pipeline processes an input image through three sequential stages:

1. **Skin Validation:** A binary classifier that determines whether the image contains valid skin. This filters out non-dermatological images early.
2. **Eczema Detection:** A binary classifier that determines whether the validated skin image shows signs of eczema.
3. **Severity Classification:** A three-class classifier that grades the detected eczema as mild, moderate, or severe.

We use the DermNet dataset from Kaggle [18] downloaded via `kagglehub` as our primary data source.

Table 1: Dataset breakdown per pipeline stage.

| Stage | Task | Classes | Image Count |
|---|---|---|---|
| 1. Skin Validation | Skin vs. Non-skin | invalid (2400), allSkin (2612) | 5012 |
| 2. Eczema Detection | Eczema vs. Clear | clearSkin (1204), eczema (1408) | 2612 |
| 3. Severity Grading | Mild / Moderate / Severe | mild (469), moderate (1885), severe (127) | 2481$^{\dagger}$ |

$^{\dagger}$After heuristic keyword labeling. Original outline reported 1407; expanded count reflects broader folder coverage across both DermNet train and test splits.

### 3.2 Stage 3 Label Construction

**Important disclaimer:** DermNet does not provide clinician-graded severity scores. Stage 3 labels are heuristically assigned by matching folder names and filenames against severity keywords:

- **Severe** — *severe, infected, weeping, crusted, erythroderma, inflamed, acute, widespread*
- **Mild** — *mild, patch, xerotic, dry, nummular, contact*
- **Moderate** — default for all remaining images

This introduces label noise: images with no distinguishing keywords are assigned moderate regardless of clinical appearance. A 70/15/15 stratified split is used (train = 1736, val = 372, test = 373).

### 3.3 Baseline Implementation

For our baseline, we use MobileNetV2 as a frozen feature extractor with global average pooling and a single Dense output layer. Images are resized to $224 \times 224$ and preprocessed with MobileNetV2's `preprocess_input` ($[-1, 1]$ scaling). We train with Adam (`lr=1e-3`) and `sparse_categorical_crossentropy` loss. Stages 1 and 2 achieve above 96% accuracy. Stage 3 with a frozen backbone achieves only ≈61% test accuracy, exhibiting a large train/test gap due to the small, imbalanced severity dataset and the absence of regularization or fine-tuning.

### 3.4 Proposed Improvements

**Architecture Upgrade.** We replace MobileNetV2 with EfficientNetB0 [8] and fine-tune the top 30 backbone layers. We integrate CBAM [7] after the backbone to focus attention on clinically relevant regions such as erythema, scaling, and lichenification. *Critical implementation note:* EfficientNetB0 applies its own internal normalization; inputs must remain in raw $[0, 255]$ — dividing by 255 causes class collapse. To ensure Keras 3 compatibility, CBAM's spatial reduce operations are implemented as named custom layers (`ReduceMeanChannel`, `ReduceMaxChannel`) rather than `Lambda` layers, which fail to serialize and cannot be cloned during Grad-CAM++ computation.

**Data Strategy.** Standard augmentations are applied during training: random horizontal and vertical flips, 90° rotations, brightness jitter (delta = 38.0 for unnormalized inputs), and contrast jitter (×0.8–1.2). EarlyStopping (patience = 7) and ReduceLROnPlateau (factor = 0.5, patience = 3) are applied throughout. HAM10000 supplementation was planned but not executed.

**Training Protocol.** All Stage 3 variants follow a two-phase procedure:

1. Phase 1: Backbone frozen, classification head trained for up to 25 epochs (`lr=1e-3`).
2. Phase 2: Top 30 backbone layers unfrozen, fine-tuned for up to 20 epochs (`lr=1e-5`).

**Explainability.** Grad-CAM++ [11] visualizations are generated using `tf-keras-vis` for one correctly classified test example per severity class, confirming the model attends to lesion regions rather than background.

## 4 Experiments

### 4.1 Experiment 1: Severity Classification Accuracy

**Goal:** Determine whether the proposed architecture and data improvements can significantly improve severity classification beyond the baseline.

We conduct an ablation study isolating the contribution of each change:

Table 2: Ablation study results for Stage 3 severity classification.

| Variant | Description | Fine-tune | Attention | Test Acc | Val Acc | Macro F1 |
|---|---|---|---|---|---|---|
| A | MobileNetV2 (baseline) | Yes | No | 81.0% | 79.0% | 0.47 |
| B | EfficientNetB0 | Yes | No | 79.9% | 80.7% | 0.55 |
| C | EfficientNetB0 + CBAM | Yes | Yes | **85.3%** | 80.4% | **0.62** |

Metrics: macro-averaged F1 is the primary metric (weights all classes equally).
All variants use the same 70/15/15 split and two-phase training protocol.

Per-class results for the best-performing variant (C) are shown below:

Table 3: Per-class results for Variant C (EfficientNetB0 + CBAM), test set.

| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Mild | 0.83 | 0.68 | 0.74 | 71 |
| Moderate | 0.88 | 0.94 | 0.91 | 283 |
| Severe | 0.25 | 0.16 | 0.19 | 19 |
| Macro avg | 0.65 | 0.59 | 0.62 | 373 |
| Weighted avg | 0.84 | 0.85 | 0.84 | 373 |

**Key findings:**

- CBAM (Variant C) improves mild recall from 0.39 (baseline) to 0.68, suggesting the attention module helps localize subtle mild-eczema texture.
- The severe class has low recall (0.16) across *all* variants. With only 19 test samples this is primarily a data size and label quality problem, not an architecture problem.
- EfficientNetB0 alone (Variant B) does not clearly dominate MobileNetV2 in accuracy (79.9% vs. 81.0%), but achieves higher macro F1 (0.55 vs. 0.47). The CBAM module drives the largest single gain.
- Planned Variant D (backbone + attention + HAM10000 augmented data) was not completed. Multi-seed variance reporting was also not performed.

### 4.2 Experiment 2: Explainability Assessment

**Goal:** Evaluate whether the improved model produces more clinically meaningful visual explanations than the baseline.

We generate Grad-CAM++ heatmaps for the best-performing variant (EfficientNetB0 + CBAM) using one correctly classified example per severity class. Figure 1 shows the results.

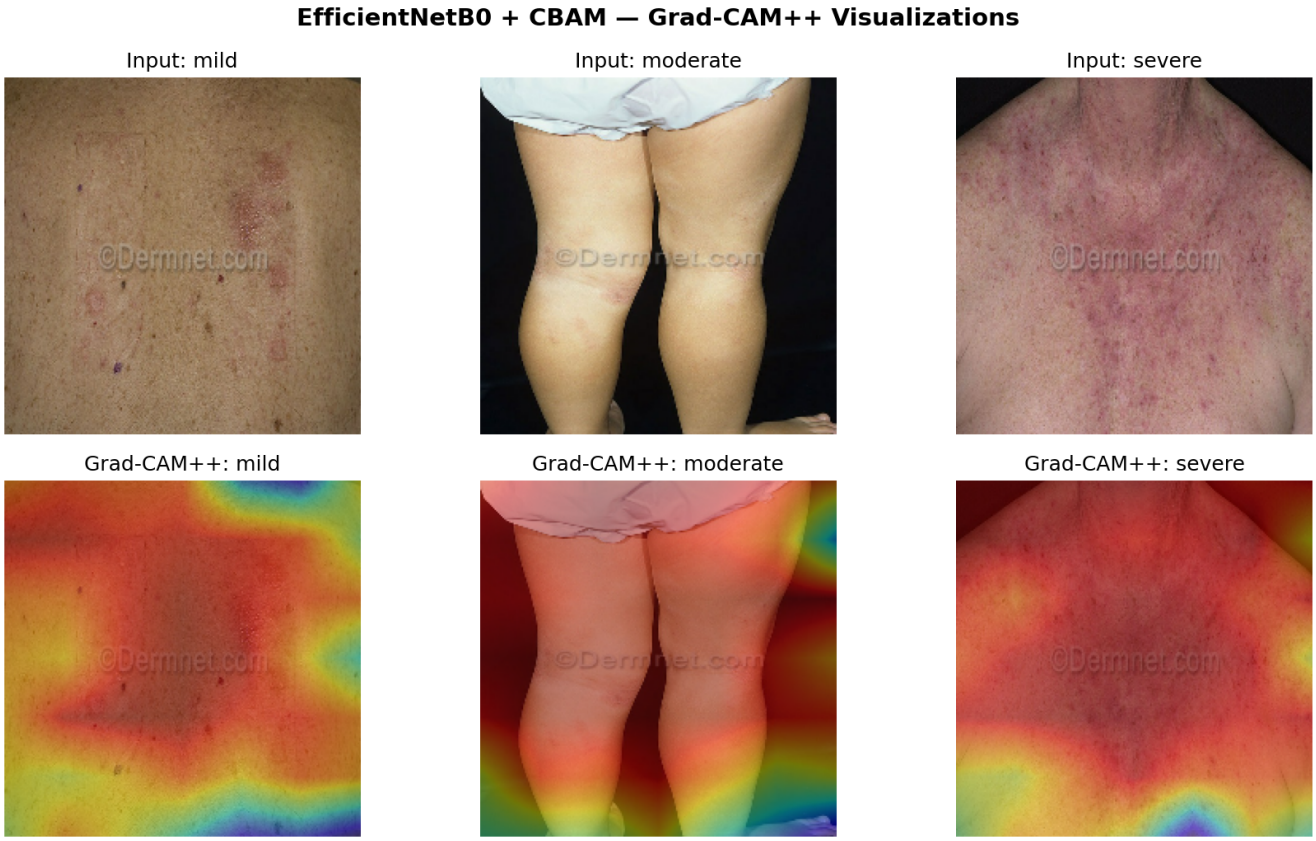


Figure 1: Grad-CAM++ visualizations for EfficientNetB0 + CBAM. For *mild* eczema, attention is distributed across dry, patchy skin texture. For *moderate*, it focuses on the inflamed knee region. For *severe*, it concentrates on the most intensely erythematous areas.

The heatmaps suggest the model responds to lesion-relevant visual cues. Quantitative evaluation using Average Drop (%) and Average Increase (%) [11], and heatmap-to-lesion IoU, was planned but not completed due to the absence of ground-truth lesion segmentation masks in the DermNet dataset.

## Discussion and Limitations

**What worked.** The three-stage pipeline cleanly separates orthogonal classification decisions. Stages 1 and 2 achieve near-perfect accuracy with simple frozen transfer learning. CBAM meaningfully improves Stage 3 macro F1 (+0.15 over baseline) and mild-class recall (+0.29).

**Limitations.**

1. **Non-clinician labels.** Severity grades are keyword-derived, not physician-assigned. The system cannot be used for clinical decision support.
2. **Small severe class.** Only 19 severe test samples makes recall estimates highly variable.
3. **Single dataset.** All data comes from DermNet; cross-dataset generalization is unknown.
4. **No quantitative explainability metrics.** Average Drop/Increase and lesion IoU were not computed.

5. **Single random seed.** Variance across training runs is unreported.

## Conclusion

We built a three-stage CNN pipeline for eczema severity classification and demonstrated that EfficientNetB0 + CBAM yields consistent improvements over a MobileNetV2 baseline: +4.3% test accuracy and +0.15 macro F1. Grad-CAM++ visualizations confirm the model attends to lesion-relevant skin regions for all three severity classes. The primary bottleneck is label quality: a clinician-annotated severity dataset would be the most impactful path to improving this system.

Code and training notebook:

https://github.com/AruruGunabhiram/attention-enhanced-eczema-severity-classification

## Acknowledgments

Dataset: DermNet via Kaggle (`shubhamgoel27/dermnet`). This project extends the baseline by Biswarup [19]. CBAM based on Woo et al. [7]. Grad-CAM++ based on Chattopadhay et al. [11].

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